

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI, HYDERABAD CAMPUS
FIRST SEMESTER 2026-27
Course Handout (Part-II)

Date: August 03, 2026

In addition to Part-I (General Handout for all courses appended to the timetable), this portion gives further specific details regarding the course.

Course No. : ECE F314
Course Title : Electromagnetic Fields and Microwave Engineering
Instructor-in-charge : Prof. Pankaj Arora

Course Description:

Review of Maxwell's equations; Wave propagation: Normal & Oblique incidence; Transmission Lines & Smith Chart, Waveguide: Rectangular & Cylindrical waveguides; Dielectric Waveguides; Cavity resonator, equivalent Circuit Model, characteristics; Microwave devices (Waveguide based and planar); S parameters; Measurement & Impedance Matching; Microwave Generators & Amplifiers; Antennas and its characteristics and types.

Scope and objective of the course:

This course provides a comprehensive understanding of electromagnetic wave theory and microwave engineering, focusing on Maxwell's equations, wave propagation, and transmission line theory. It covers the analysis and design of waveguides, cavity resonators, and dielectric structures. Students will explore microwave devices, S-parameters, impedance matching techniques, and measurement methods. The course also introduces microwave generators, amplifiers, and various types of antennas with their characteristics. The objective is to equip students with the theoretical foundation and practical skills needed to analyze, design, and apply high-frequency systems in modern communication, radar, and sensing applications.

Learning Outcomes:

Upon completion of this course, students will be able to analyze electromagnetic wave propagation using Maxwell's equations, understand wave behavior at interfaces, and design transmission lines with impedance matching using the Smith Chart. They will gain proficiency in characterizing waveguides, cavity resonators, and microwave devices, as well as interpreting S-parameters for network analysis. Students will also learn measurement techniques, explore microwave sources and amplifiers, and evaluate various antenna types and their radiation characteristics for practical applications.

1. Textbook:

- 1) TB1: David Pozar, "Microwave Engineering", 4th edition, John Wiley & Sons, 2012.
- 2) TB2: John D. Kraus and Daniel A. Fleisch, "Electromagnetics", 5th ed., McGraw-Hill, New York, 1999.

2. Reference Books :

- i) RB1: David K. Cheng, “Field and Wave Electromagnetics,” 2nd ed. Pearson Education, New Delhi, 2009.
- ii) RB2: Samuel Y. Liao, “Microwave devices and circuits,” 3rd ed. , PHI 2008.
- iii) RB3: J. D. Krauss et al., “Antennas and Wave Propagation”, 4th edition, TMH 2010.

3. Course Plan:

Lec. No.	Topic to be covered	Learning Objective	Reference
1	Introduction to the course and its components	Basic Introduction	--
2-4	Transmission lines	To analyze the transmission line characteristics	Class notes and TB1
5-8	Graphical Evaluation of Transmission Lines with Smith Chart	To analyze transmission lines and high-frequency systems using a graphical technique (Smith Chart)	Class notes and TB1
9	Review of Maxwell Equations	Review of basic EM concepts	Class notes and TB2
10-12	Wave propagation in conducting and dielectric media with normal & Oblique incidence	To qualitatively and quantitatively describe the propagation of EM waves in different media	Class notes and TB2
13-15	S parameters, measurement, and impedance matching	To quantitatively analyze the power in EM waves	Class notes and TB2
16-18	Reflection & refraction of plane waves	To describe various wave phenomena quantitatively	Class notes and TB2
19-22	Waveguides & Cavity resonators	To quantitatively analyze waveguide propagation	Class notes and TB1
23-27	Microwave devices (Waveguides and planar-based)	To qualitatively describe various microwave devices	Class notes and TB1
28-35	Microwave Generators and Amplifiers	To qualitatively and quantitatively analyze low and high-power microwave generators and amplifiers	Class notes and RB2
36-40	Antennas principles, radiating elements & types	To quantitatively and qualitatively analyze RF and microwave antennas	Class notes and RB3

4. Evaluation Scheme:

Component	Duration	Percentage	Marks	Nature of Component
Mid-sem Test	90 min	30 %	30	Closed Book
Quizzes	TBA	30 %	30	Closed Book

Comprehensive. Exam.	180 min	40 %	40	Closed/Open Book
Total			100	

5. **Chamber Consultation Hour:** To be announced in class
6. **Notices:** All notices concerning the course will be put up on the Nalanda.
7. **Minimum performance criterion:** A student should secure 25 % of the class average, or 10 % of the maximum mark, whichever is lower, to clear the course with a letter grade.
8. **Make-up Examination:**
No make-up will be given for Quizzes in any case. Makeup for mid-semester and Comprehensive examinations will be only in **extremely genuine cases**.

Prof. Pankaj Arora
INSTRUCTOR-IN-CHARGE
ECE F314